

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	xxxxxx
Project Name	PocketSmart AI
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

PocketSmart AI is a Generative AI-powered budget optimization and recommendation assistant. It helps users plan purchases or events within a defined budget and reduces the effort required to search, compare and shortlist suitable options.

Project Overview	
Objective	To help users make faster and more confident budget-aware decisions by combining user preferences, budget constraints and AI-powered recommendations.
Scope	The solution covers three initial use cases: Home Interior Planning, Party Planning, and Jewelry Recommendations. It supports budget entry, preference capture, recommendation generation, budget-aware allocation and comparison.
Problem Statement	Users often face too many choices across products and services, making it difficult and time-consuming to compare options while staying within a fixed budget.
Impact	Reducing manual comparison and presenting personalized options can make planning easier, reduce the risk of overspending, and improve user confidence in the final decision.
Proposed Solution	
Approach	Use a recommendation workflow that accepts the user's category, budget and preferences, processes the constraints, generates personalized suggestions, and presents a shortlist with estimated spending and comparison information.
Key Features	- Budget-aware recommendations - Personalized suggestions based on category and preferences - Budget allocation / optimization - Comparison of shortlisted options - Support for Home Interior, Party Planning and Jewelry - Ability to refine preferences and regenerate results

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing resources	Standard development laptop/desktop; optional GPU for AI model experimentation if required.
Memory	RAM specifications	8 GB or higher recommended for development.
Storage	Data, models and logs	At least 50 GB available for prototype development; expandable based on datasets and model artifacts.
Software	Programming / framework	Python-based AI/backend services; React-based web client for the prototype.
Libraries / Tools	AI, data and development libraries	Generative AI SDK/API, pandas, NumPy and relevant ML/data-processing libraries as required.
Development Environment	IDE / collaboration	VS Code or Jupyter Notebook; Git/GitHub for source control.
Data / External Sources	Product/service information	External product/service APIs or data sources where available; recommendation inputs supplied by users.